

D848

General Earth Science I Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A lab group using a stream table donated by a local museum changes the slope of the table (5°, 10°, 15°) and measures how much sand is carried away in 2 minutes. They use the same water flow and sand type each time. The group's table is painted blue. What is the dependent variable?

- A. The water flow.
- B. The slope of the table.
- C. The amount of sand carried away.
- D. The color of the table.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A student wants to test whether water flow rate affects erosion. Which variable should be held constant?

- A. The amount of sand eroded.
- B. Nothing needs to be held constant.
- C. The slope of the stream table.
- D. The water flow rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Students test whether a plant cover reduces soil erosion. Three trays have grass cover, and one tray has bare soil. All trays get the same soil, slope, and water. The class did the lab on a sunny Friday. What is the bare-soil tray?

- A. The control group.
- B. An error, because all trays should be the same.
- C. A controlled variable.
- D. The independent variable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

During a rock-testing lab in a room with new lab benches, a student splashes dilute acid in her eye while testing limestone. Her partner is recording data on a tablet. What should happen first?

- A. Wipe the eye with a paper towel.
- B. Flush her eye at the eyewash for at least 15 minutes and alert the instructor.
- C. Finish recording the data first.
- D. Clean up the spill on the bench.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A seismograph at a station in a quiet farming town records a P wave and then an S wave 40 seconds later. The class uses the rule that each second of S-P lag equals 8 km. The station is 20 years old. How far is the station from the epicenter?

- A. 160 km
- B. 320 km
- C. 5 km
- D. 48 km

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Using the same rule (8 km per second of S-P lag), a station is 480 km from an epicenter. What S-P interval should it record?

- A. 472 seconds
- B. 3,840 seconds
- C. 0 seconds, since P and S waves travel together.
- D. 60 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A student uses a free online erosion simulation with colorful 3-D graphics. It shows a riverbank eroding 2 m a year. She writes that a real river near her school will erode exactly 2 m a year. The simulation has thousands of users. What is the problem?

- A. The simulation simplifies reality; its result must be compared with field data for that river.
- B. None; simulations are always accurate.
- C. The graphics should be more realistic.
- D. None; thousands of users confirm it.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

At 6 p.m. a weather station at a school with a rooftop garden reads an air temperature of 20°C and a dew point of 12°C. Overnight, the air cools to 12°C. The station was installed last spring. What will most likely happen?

- A. The rooftop garden prevents condensation.
- B. Water freezes, since dew point is the freezing point.
- C. Nothing, since 12°C is above freezing.
- D. Dew or fog forms because the air is saturated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

The air temperature equals the dew point. What is the relative humidity?

- A. Cannot be known without the temperature value.
- B. 50%
- C. 100%
- D. 0%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A class with a new weather balloon finds that this July's average temperature was 31°C, while the 30-year July normal is 28°C. The data came from a website with an easy-to-read layout. What can they conclude?

- A.** The climate has warmed by 3°C.
- B.** This July was 3°C warmer than normal; one year alone does not show a climate trend.
- C.** The data are wrong because the site is easy to read.
- D.** This July was 3°C cooler than normal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a lab report typed in a clean font, a student writes: The table shows sand loss increased from 40 g to 95 g as slope increased, which suggests steeper slopes speed erosion. The report is four pages long. Which section is this sentence best suited for?

- A.** Hypothesis.
- B.** Materials.
- C.** Procedure.
- D.** Analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Which sentence belongs in the conclusion of a lab report?

- A.** The data supported the hypothesis that steeper slopes increase erosion.
- B.** If slope increases, then erosion will increase.
- C.** Pour 500 mL of water down the stream table.
- D.** Trial 3: 95 g of sand lost.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A lab group using a stream table ~~donated by a local museum~~ changes the slope of the table (5°, 10°, 15°) and measures how much sand is carried away in 2 minutes. They use the same water flow and sand type each time. ~~The group's table is painted blue.~~ What is the dependent variable?

Target word: dependent variable **Decides:** What is measured.

A	Adjacent category	A controlled variable.
B	Inverted operation	That is the independent variable.
C	Correct	The measured outcome.
D	Camouflage	Irrelevant.

2. Answer C Concept 1 • transfer

A student wants to test whether water flow rate affects erosion. Which variable should be held constant?

Target word: identify a controlled variable **Decides:** Everything except flow rate and the measured erosion stays the same.

A	Confusion set	That is the dependent variable.
B	Absolute language	Uncontrolled variables confound results.
C	Correct	Must stay the same so only flow changes.
D	Inverted operation	That is the independent variable.

3. Answer A Concept 2

Students test whether a plant cover reduces soil erosion. Three trays have grass cover, and one tray has bare soil. All trays get the same soil, slope, and water. ~~The class did the lab on a sunny Friday.~~ What is the bare-soil tray?

Target word: role of bare tray **Decides:** It receives no treatment.

A	Correct	An untreated baseline for comparison.
B	Inverted operation	The comparison is intended.
C	Confusion set	Controlled variables are conditions held constant.
D	Adjacent category	Plant cover is the independent variable.

4. Answer B Concept 3

During a rock-testing lab ~~in a room with new lab benches~~, a student splashes dilute acid in her eye while testing limestone. ~~Her partner is recording data on a tablet.~~ What should happen first?

Target word: first safety response **Decides:** Eye exposure to a chemical.

A	Adjacent category	Does not remove the chemical safely.
B	Correct	Immediate flushing protects the eye.
C	Camouflage	Data never comes before safety.
D	Inverted operation	Cleanup comes after the person is safe.

5. Answer B Concept 4

A seismograph at a station ~~in a quiet farming town~~ records a P wave and then an S wave 40 seconds later. The class uses the rule that each second of S-P lag equals 8 km. ~~The station is 20 years old.~~ How far is the station from the epicenter?

Target word: compute distance **Decides:** $40 \text{ s} \times 8 \text{ km/s}$.

A	Camouflage	Multiplies by the station age instead of the lag rule (20×8).
B	Correct	$40 \times 8 = 320 \text{ km}$.
C	Inverted operation	$40 / 8$.
D	Wrong stem number	$40 + 8$.

6. Answer D Concept 4 • transfer

Using the same rule (8 km per second of S-P lag), a station is 480 km from an epicenter. What S-P interval should it record?

Target word: solve for lag **Decides:** $480 / 8$.

A	Wrong stem number	$480 - 8$.
B	Inverted operation	Multiplies instead of dividing.
C	Confusion set	P waves are faster.
D	Correct	$480 / 8 = 60 \text{ s}$.

7. Answer A Concept 5

A student uses a free online erosion simulation ~~with colorful 3-D graphics~~. It shows a riverbank eroding 2 m a year. She writes that a real river near her school will erode exactly 2 m a year. ~~The simulation has thousands of users.~~ What is the problem?

Target word: limit of simulation **Decides:** Simulation results need real-data checks.

A	Correct	Models have assumptions and limits.
B	Absolute language	They simplify.
C	Adjacent category	Graphics do not change the model's assumptions.
D	Camouflage	Popularity is not validation.

8. Answer D Concept 6

At 6 p.m. a weather station ~~at a school with a rooftop garden~~ reads an air temperature of 20°C and a dew point of 12°C . Overnight, the air cools to 12°C . ~~The station was installed last spring.~~ What will most likely happen?

Target word: what happens at dew point **Decides:** Air cools to its dew point.

A	Camouflage	Irrelevant.
B	Confusion set	Dew point is not the freezing point.
C	Dropped element	Ignores saturation.
D	Correct	At the dew point, relative humidity is 100%.

9. Answer C Concept 6 • transfer

The air temperature equals the dew point. What is the relative humidity?

Target word: reverse: from condition to humidity **Decides:** Saturation.

A	Dropped element	Equality alone shows saturation.
B	Adjacent category	No basis.
C	Correct	Air at its dew point is saturated.
D	Inverted operation	The opposite of saturated.

10. Answer B Concept 7

A class ~~with a new weather balloon~~ finds that this July's average temperature was 31°C, while the 30-year July normal is 28°C. ~~The data came from a website with an easy to read layout.~~ What can they conclude?

Target word: what the data support **Decides:** One year compared with a normal.

A	Absolute language	One year cannot define climate.
B	Correct	An anomaly, not a trend.
C	Camouflage	Irrelevant.
D	Inverted operation	31 is above 28.

11. Answer D Concept 8

In a lab report ~~typed in a clean font~~, a student writes: The table shows sand loss increased from 40 g to 95 g as slope increased, which suggests steeper slopes speed erosion. ~~The report is four pages long.~~ Which section is this sentence best suited for?

Target word: choose section **Decides:** It interprets the data.

A	Inverted operation	The hypothesis predicts before data.
B	Confusion set	Lists equipment.
C	Adjacent category	Procedure lists steps.
D	Correct	Interprets the data.

12. Answer A Concept 8 • transfer

Which sentence belongs in the conclusion of a lab report?

Target word: produce a conclusion sentence **Decides:** Answers the question and evaluates the hypothesis.

A	Correct	Evaluates the hypothesis.
B	Inverted operation	A hypothesis, stated before data.
C	Adjacent category	A procedure step.
D	Dropped element	Raw data without interpretation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Independent, dependent, and controlled variables	The independent variable is what the experimenter changes; the dependent variable is what is measured; controlled variables are kept the same.	Mixing up the variable changed with the variable measured.	An experiment description asking which variable is which.
2	Control group vs. controlled variables	A control group receives no treatment and serves as a baseline; controlled variables are conditions held constant across all groups.	Using the two terms interchangeably.	A setup with an untreated comparison.
3	Lab safety response	For a chemical splash to the eye, flush at the eyewash for at least 15 minutes and alert the instructor; safety response comes before cleanup or data.	Finishing the procedure or cleaning up first.	An accident during a lab and a choice of first action.
4	S-P interval and distance	The time between P-wave and S-wave arrivals increases with distance from the epicenter; a travel-time relationship converts lag to distance.	Thinking a longer lag means a closer or stronger quake.	A seismogram with P and S arrival times.
5	Limits of simulations	A simulation simplifies reality; its results should be checked against real data and its assumptions stated.	Treating simulation output as equal to field observation.	A virtual lab result used to make a real-world claim.
6	Dew point and saturation	When air cools to its dew point, it is saturated (100% relative humidity) and condensation forms dew, fog, or clouds.	Believing dew point is the temperature at which water freezes.	An air temperature and a dew point in the same question.
7	Climate needs long-term data	Climate is judged from long-term averages, such as 30-year normals; one year is an anomaly, not a trend.	Declaring a climate trend from one year's data.	One year compared with a long-term normal.
8	Parts of a lab report	The hypothesis predicts; the procedure lists steps; the data/results report observations; the analysis interprets; the conclusion answers the question and evaluates the hypothesis.	Putting interpretation in results or new data in the conclusion.	A sentence that must be placed in a report section.

Course scope (public description): The Earth Science I Lab course equips students with skills to conduct scientific investigations in Earth science. Students will apply the scientific method, design controlled experiments, and follow lab safety protocols. They will gain experience with data collection methods, including field observations, experiments, and virtual simulations of phenomena like erosion. The course emphasizes analyzing weather, climate, and seismic data using basic techniques. It also covers essential aspects of identifying and analyzing the components of a lab report by answering questions related to a simulated lab report. Students will learn to interpret data, draw conclusions, and communicate findings effectively, preparing them for advanced Earth science studies and research.